Student Answer Sheet

Subject: Artificial Intelligence

Definition:

Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, especially computer systems. These processes include learning (the acquisition of information and rules for using it), reasoning (using rules to reach approximate or definite conclusions), and self-correction. AI enables machines to perform tasks that typically require human intelligence, such as visual perception, speech recognition, decision-making, and language translation.

Body:

AI applications span multiple domains. Machine Learning (ML) is a subset of AI where systems learn from data without being explicitly programmed. Deep Learning uses neural networks with many layers to recognise patterns. Natural Language Processing (NLP) enables machines to understand and generate human language. Computer Vision allows machines to interpret visual information. Robotics integrates AI for autonomous physical actions. Key AI techniques include supervised learning, unsupervised learning, reinforcement learning, and transfer learning. AI is used in healthcare for diagnostics, in finance for fraud detection, in transportation for self-driving cars, and in education for personalised learning systems.

Conclusion:

Artificial Intelligence represents a transformative technology that is reshaping industries and society. While AI offers immense benefits in automation, efficiency, and problem-solving, it also raises ethical concerns about privacy, bias, job displacement, and autonomous decision-making. Responsible AI development requires transparency, fairness, and accountability. The future of AI depends on advancing research in explainable AI, general AI, and quantum AI while ensuring human-centric design principles.